

WRO (Future Engineers)

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Introduction

This competition is designed for students aged 14-22, where teams of 2-3, guided by a coach, get the opportunity to work on real-world engineering challenges. You'll design, build, and program a robot that can drive autonomously on a track, using any hardware or software you choose—within a size limit of 30 x 20 x 30 cm. It's a chance to explore advanced robotics, learn about the full engineering workflow, and document your work on GitHub. In the WRO Future Engineers category teams need to focus on all parts of the engineering process.

Who Can Participate?

Age Range: 14 – 22 years (new starting 2025 season)

Team Structure:

- Teams of 2-3 students
- Each team is guided by a coach

What is WRO® Future Engineers?

A category designed for older students, combining current research trends with robotics

Aims to bring real-world research challenges into schools

Focus on the complete engineering workflow:

- Design
- Build
- Program
- Document on GitHub



The Challenge: Autonomous Driving

Objective: Build a robot that drives autonomously on a track

Key Feature: Must have a steering drive (choose any robot, controller, and materials in line with the rules)



Technical Requirements

Hardware: Free choice – build your own robot with any hardware

Software: Free choice – select any programming language or controller

Size Constraint: Maximum dimensions of 30 x 20 x 30 cm

Encourages innovation and creativity within set limits

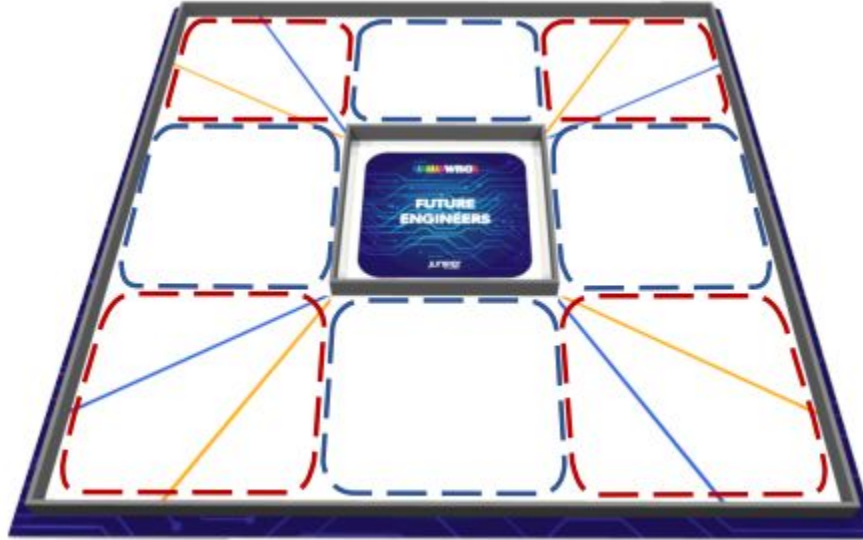
Learning Outcomes & Skills

1. Engineering workflow and project management
2. Teamwork and problem solving
3. Real-world robotics challenges like sensor fusion and computer vision
4. Creating detailed engineering documentation on GitHub

How the Competition Works

- Two main rounds: Open Challenge & Obstacle Challenge
- Autonomous driving on a variable track layout
- Focus on both vehicle performance and documentation (approx. 75% for performance, 25% for documentation)
- Follow detailed rules (all specifics in the General Rules document):
<https://wro-association.org/wp-content/uploads/WRO-2025-Future-Engineers-Self-Driving-Cars-General-Rules.pdf>

How the Competition Works





Why Participate?

- Gain hands-on experience with advanced robotics and engineering design
- Work collaboratively in teams under the guidance of experienced coaches
- Enhance your resume and technical skills through real-world challenges
- Opportunity to innovate and learn from current research trends

How to Get Started

1. Visit the WRO website for detailed guidelines and rule documents:
<https://wro-association.org/competition/2025-season/#rules>
2. Form your team and find a coach who can guide you.
3. Start brainstorming ideas for your autonomous driving robot.
4. Begin preparing your engineering documentation early.

Questions

Feel free to ask me any questions you have